

The Conversion Engine — Decision Memo

To: Tenacious CEO & CFO **From:** Chalie Lijam · TRP1 Week 10 **Date:** 2026-04-25 **Repo:** The_Conversion_Engine / main

1. EXECUTIVE SUMMARY

The Conversion Engine replaces ad-hoc LinkedIn prospecting with a systematic pipeline that enriches every B2B prospect from three public data sources, classifies them into one of four ICP segments with a confidence-gated LLM email, and syncs every event to HubSpot in real time. On Tenacious-specific failure probes it significantly reduces the two highest-cost failure modes — bench over-commitment (−42pp, $p<0.001$ C−04) and tone non-compliance (−30pp, $p=0.003$ C−05) — while holding cost per qualified lead at \$0.015, well below the \$5 hard cap C−06 . τ^2 -Bench retail held-out pass@1 is 0.70 vs baseline 0.7667 (−6.7pp, n.s.) — honest result reported; see §2. We recommend a 30-day pilot on Segment 1 (recently funded Series A/B) at 50 leads/week before expanding to further segments.

2. PERFORMANCE BASELINE

Metric	Day-1 Baseline	Full Method	Delta	Significance
τ^2 -Bench retail pass@1 (dev, n=150, staff baseline)	0.7267 [0.6504,0.7917] C−01	—	—	Reference only
τ^2 -Bench retail pass@1 (held-out, n=30, 1 trial, Claude Sonnet 4.6)	0.7667 C−02	0.7000 C−03	−6.7pp	$p=0.72$ — n.s. (underpowered, need $n\approx440$)
Bench compliance rate (P-009 probe)	0.30	0.72	+42pp	$p<0.001$ *** C−04
Tone compliance rate	0.61	0.91	+30pp	$p=0.003$ ** C−05
Cost per qualified lead	\$0.0199/sim C−07	\$0.015	−25%	C−06

Real evaluation: Claude Sonnet 4.6 via OpenRouter, 30 held-out tasks, 1 trial each (2026-04-25). τ^2 -Bench retail measures general B2B agent capability; it does not exercise Tenacious-specific constraints (bench gating, offshore tone). The −6.7pp held-out result is expected and honest — the method adds constraints irrelevant to generic retail tasks. Tenacious-specific probe metrics (+42pp bench compliance, +30pp tone) are the primary significance measures.

3. COST PER QUALIFIED LEAD

\$0.015/lead — \$0.012 LLM compose · \$0.002 tone guard · \$0.0006 expected regeneration (30% rate). C−06
At the \$4 challenge-week dev budget: 267 leads addressable. Full production at 200 leads/week: \$156/year LLM cost.

4. STALLED-THREAD RATE DELTA

Current Tenacious (manual): 30–40% of qualified conversations stall in the first two weeks due to partner context-recall delay (CEO/CFO estimate). **System proxy:** thread context (hiring brief + full history) is surfaced in <1s on any inbound webhook event vs. an estimated 2–4h partner recall delay. Assuming context delay drives 50% of manual stalls: projected system stall rate $\leq 20\%$. C−15

No live stall data available — kill-switch was active during challenge week. Confirmation requires 30-day pilot with real prospect replies.

5. COMPETITIVE-GAP OUTBOUND PERFORMANCE

Variant	Signal injection	Projected reply rate
Full method (non-abstained)	100% include AI maturity score + top-quartile gap C−16	7–12%
Baseline (generic pitch)	0%	2–4%
Abstained (confidence <0.40)	Generic exploratory only · ~18% of pipeline	—

Reply rates are industry benchmarks for signal-grounded vs. generic cold B2B email (Woodpecker 2024). Live reply delta unmeasurable during challenge week (kill-switch active).

6. ANNUALIZED DOLLAR IMPACT

Scenario	Leads/week	Replies (8%)	Meetings (40%)	Deals (15%)	Revenue @ \$36K ACV	LLM cost/yr
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Segment 1 only (Series A/B)	50	208	83	12	\$432K C-17	\$39
Segments 1 + 3 (+ leadership)	100	416	166	25	\$900K C-18	\$78
All 4 segments	200	832	333	50	\$1.8M C-19	\$156

ACV \$36K from internal Tenacious estimates (Risk 4 derivation). Conversion rates are conservative industry benchmarks. Does not include SDR labour cost for meeting follow-through.

7. PILOT SCOPE RECOMMENDATION

Segment: Segment 1 — Recently Funded Series A/B (clearest signal, highest classifier confidence, lowest offshore-perception risk).
Volume: 50 leads/week × 4 weeks = 200 leads total. SDR reviews each draft before send (~2h/week).
Budget: \$3 LLM cost + SDR time. Kill-switch remains unset until SDR approval gate is wired.
Success criterion: ≥8 qualified discovery calls booked in 30 days (4% meeting rate — conservative below expected 6.4%). If met: expand to Segment 3 and raise weekly volume to 100.

8. FOUR FAILURE MODES T²-BENCH DOES NOT CAPTURE

Failure Mode	Specific trigger	Business impact	Status
Bench over-commitment	Agent commits "3 Go engineers available" when bench_summary.json is 48–72h stale and the stack is fully engaged	\$44K/event expected C–10	Reduced 7→2/10. Not eliminated.
Offshore-perception language	Segment 2 pitch uses "replace higher-cost roles" framing to a CTO who has had a failed offshore engagement — triggers immediate delete or complaint	\$36K lost deal + brand signal	Tone guard catches style markers; cultural framing not tested.
ICP misclassification on conflict signals	Company had layoff 90d ago AND new CTO 45d ago; agent wavers between Segment 2 and Segment 3, sends a message referencing both — signals feel researched-but-confused C–20	\$18K/event	P-004 residual 3/10. Unresolved. See §12.
Multi-thread leakage	Two Tenacious SDRs contact CEO and CTO of the same company in the same week with different segment pitches; prospect receives contradictory framing	\$108K/event (deal killed)	P-016 0/10 in probe. Thread store enforces one active thread per company.

9. PUBLIC-SIGNAL LOSSINESS

False negative — Quiet-sophisticated

A financial-services firm with a proprietary trading ML system that keeps its stack and team private (no AI job titles, no GitHub org, no public exec commentary). AI maturity score: 0–1. Agent either abstains or sends a generic opener. Tenacious misses a potential Segment 4 high-ACV deal to a competitor with a warm introduction. Estimated false-negative rate: ~15% of true Segment 4 companies are publicly silent.

False positive — Loud-but-shallow

A startup that posts "AI Engineer" and "ML Platform Lead" as aspirational roles with no actual ML infrastructure — the titles are recruiting theatre. Score: 2–3. Agent sends a Segment 4 capability pitch. The CTO knows they have no ML maturity and reads the email as tone-deaf overreach, deleting it or filing a complaint. Signal: job titles without ML stack evidence should reduce, not anchor, confidence.

10. GAP-ANALYSIS RISKS

Top-quartile benchmark irrelevant for product-embedded AI. The gap brief flags "no Named Head of AI" as a leadership gap. For a SaaS company whose product is AI (every engineer works on the model), this gap is a deliberate organisational choice, not a deficiency. The agent would pitch Tenacious as filling a gap that the prospect considers already solved internally. The signal in our data: DataSync AI's own gap brief flagged this pattern — CTO is ex-Google Brain but there is no separate "Head of AI" role. The pitch would be wrong for this company as a Segment 4 target.

Modern ML stack benchmark wrong for OT/IoT manufacturing. The top-quartile benchmark assumes cloud-native Snowflake + MLflow stacks. A discrete-manufacturing prospect whose bottleneck is OT sensor integration, not ML pipelines, would receive a pitch about data infrastructure that is architecturally irrelevant to their problem. The agent has no sub-niche discriminator for hardware-adjacent AI. Business impact: a well-researched but wrong pitch accelerates a "not relevant to us" response.

11. BRAND-REPUTATION UNIT ECONOMICS

1,000 signal-grounded outbound emails sent	
Reply rate (top-quartile benchmark):	7–12% → 70–120 replies
Wrong-signal email rate assumption:	5% → 50 emails with factually wrong data
Reputation cost per wrong email:	\$500 (1 lost future referral × 0.5 ACV)
Expected brand damage:	50 × \$500 = \$25,000
Expected revenue from replies:	95 replies × 15% conversion × \$36,000 ACV = \$512,100
Net (before sales cost):	+\$487,100 C–22
Break-even wrong-signal rate:	~29% – current 5% is well within safe range

Risk materialises if signal quality degrades (e.g. layoffs.fyi false-positive rate increases) or brand damage per event exceeds \$3,000. Monitored via probe trigger rates on each campaign run.

12. ONE HONEST FAILURE — P-004 (LAYOFF + NEW CTO CONFLICT)

Probe: A company had a 15% workforce layoff 90 days ago and hired a new CTO 45 days ago. Before fix: 5/10 trials produced a conflicted email referencing both signals without resolving the tension. After fix: 3/10. C-20

Not resolved. The ICP classifier assigns the highest-confidence segment independently but the LLM system prompt does not explicitly resolve the conflict when both `layoff\_event` and `leadership\_change` are present. The agent sometimes produces copy that reads "we noticed your recent restructuring — and with your new CTO in place, this is a vendor-reassessment window", which a reader correctly identifies as the agent having two different scripts running simultaneously. **Impact if deployed:** \$18K expected deal loss per triggered event. At 3/10 trigger rate on conflict-signal prospects, this is the highest residual risk in the probe library.

13. KILL-SWITCH CLAUSE

The Tenacious CEO should pause this system if **any one** of these three conditions is met in any rolling 7-day window: C-21

- **Spam complaint rate >0.3%** — measured via Resend `email.complained` webhook events. Rationale: Resend auto-suspends at 0.3–0.5%; preemptive pause preserves sender reputation. Threshold resets after manual review of flagged emails and prompt audit.
- **P-009 probe trigger rate >3/10** on the weekly automated probe run. Rationale: indicates bench_summary.json is stale or the injection prompt has regressed. Resume only after bench data is refreshed and probe returns ≤2/10.
- **More than 3 HubSpot contacts manually flagged "wrong signal"** by SDRs in any 7-day window (via `tenacious\_status=rejected` field). Rationale: field feedback is the ground truth that public-signal lossiness has crossed a tolerable threshold. Resume only after AI maturity scorer is recalibrated against the flagged examples.

*All claims traceable via memo/evidence_graph.json (22 claims C-01–C-22) · Validator: eval/evidence_graph_validator.py ·
Charlie Lijam · 2026-04-25*